

Analysing Big Data with Hadoop

Big Data is unwieldy because of its vast size, and needs tools to efficiently process and extract meaningful results from it. Hadoop is an open source software framework and platform for storing, analysing and processing data. This article is a beginner's guide to how Hadoop can help in the analysis of Big Data.



Big Data is a term used to refer to a huge collection of data that comprises both structured data found in traditional databases and unstructured data like text documents, video and audio. Big Data is not merely data but also a collection of various tools, techniques, frameworks and platforms. Transport data, search data, stock exchange data, social media data, etc, all come under Big Data.

Technically, Big Data refers to a large set of data that can be analysed by means of computational techniques to draw patterns and reveal the common or recurring points that would help to predict the next step—especially human behaviour, like future consumer actions based on an analysis of past purchase patterns.

Big Data is not about the volume of the data, but more about what people use it for. Many organisations like business corporations and educational institutions are using this data to analyse and predict the consequences of certain actions. After collecting the data, it can be used for several functions like:

- Cost reduction
- The development of new products
- Making faster and smarter decisions
- Detecting faults

Today, Big Data is used by almost all sectors including banking, government, manufacturing, airlines and hospitality.

There are many open source software frameworks for storing and managing data, and Hadoop is one of them. It has a huge capacity to store data, has efficient data processing power and the capability to do countless jobs. It is a Java based programming framework, developed by

Apache. There are many organisations using Hadoop — Amazon Web Services, Intel, Cloudera, Microsoft, MapR Technologies, Teradata, etc.

The history of Hadoop

Doug Cutting and Mike Cafarella are two important people in the history of Hadoop. They wanted to invent a way to return Web search results faster by distributing the data over several machines and make calculations, so that several jobs could be performed at the same time. At that time, they were working on an open source search engine project called Nutch. But, at the same time, the Google search engine project also was in progress. So, Nutch was divided into two parts—one of the parts dealt with the processing of data, which the duo named Hadoop after the toy elephant that belonged to Cutting's son. Hadoop was released as an open source project in 2008 by Yahoo. Today, the Apache Software Foundation maintains the Hadoop ecosystem.

Prerequisites for using Hadoop

Linux based operating systems like Ubuntu or Debian are preferred for setting up Hadoop. Basic knowledge of the Linux commands is helpful. Besides, Java plays an important role in the use of Hadoop. But people can use their preferred languages like Python or Perl to write the methods or functions.

There are four main libraries in Hadoop.

1. **Hadoop Common:** This provides utilities used by all other modules in Hadoop.